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ANALYTICS

Foundations of Machine Learning

Project Report (Deliverable 2)

Heart Disease Prediction using Machine Learning Algorithms

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Abstract

Cardiovascular diseases (CVDs) remain the leading cause of mortality globally, necessitating effective tools for early detection and prevention. This project aims to develop a robust machine learning model to predict the presence of cardiovascular disease using demographic, lifestyle, and clinical data from 70,000 patients. We formulated the task as a binary classification problem and implemented a comprehensive pipeline including data cleaning, advanced feature engineering (e.g., Body Mass Index, Mean Arterial Pressure, Metabolic Score), and model optimization. We evaluated several algorithms, including Logistic Regression, K-Nearest Neighbors (KNN), SVM, and Gradient Boosting methods. Our final best model was an optimized XGBoost classifier, which achieved an accuracy of 73.15% on the test set with a weighted F1-score of 0.73, demonstrating strong generalization capabilities with minimal overfitting. Feature importance analysis revealed that systolic blood pressure, age, and cholesterol levels are the most critical predictors of heart disease risk. Furthermore, our custom-engineered features, particularly the composite metabolic risk score, significantly contributed to the model’s predictive power. These results highlight the potential of machine learning to assist healthcare professionals in early diagnosis, potentially reducing the burden of invasive diagnostic procedures.

1 Introduction

Cardiovascular diseases (CVDs) encompass a group of disorders affecting the heart and blood vessels, including coronary artery disease, heart failure, and stroke. They represent a critical global health challenge, currently standing as the leading cause of death worldwide. According to the World Health Organization (WHO), CVDs claimed approximately 17.9 million lives in 2019, representing nearly 32% of all global deaths [1]. Beyond the tragic loss of life, these conditions impose a colossal economic and social burden on healthcare systems, exacerbated by modern lifestyle factors such as unhealthy diets, physical inactivity, tobacco use, and chronic stress[2].

Early detection is paramount to reducing mortality rates and mitigating long-term complications. However, traditional diagnostic methods such as coronary angiography or stress tests often present significant barriers: they can be invasive, prohibitively expensive, time-consuming, and require specialized medical infrastructure not readily available in all regions. Consequently, there is an urgent need for non-invasive, cost-effective, and accessible screening tools that can identify at-risk individuals before acute cardiac events occur.

In this context, Machine Learning (ML) offers a promising avenue to revolutionize cardiovascular diagnostics. By leveraging large datasets of routine clinical and demographic data, ML algorithms can uncover complex, non-linear patterns that may elude conventional statistical analysis or human intuition [3]. This project aims to develop and evaluate a robust predictive model capable of identifying the presence of cardiovascular disease based on standard, easily obtainable medical indicators (such as blood pressure, cholesterol levels, and body mass index). Our approach focuses not only on predictive accuracy but also on model interpretability and generalization, ensuring that the proposed solution is reliable enough for potential clinical decision support.

2 Motivation & Problem Definition

2.1 Motivation

The motivation behind this study is threefold: improving preventive healthcare, enabling personalized monitoring, and supporting public health initiatives.

1. **Preventive Healthcare:** A reliable predictive model enables early screening. By analyzing simple, non-invasive metrics, healthcare professionals could identify high-risk patients during routine check-ups, facilitating early lifestyle interventions or preventive pharmacological treatments.
2. **Personalized Monitoring:** Such models can be integrated into wearable health technology. As personal health devices (smartwatches, fitness trackers) become omnipresent, embedding predictive algorithms could provide users with real-time feedback on their cardiovascular risk, fostering proactive self-care.
3. **Public Health:** On a macro scale, aggregated predictions can aid public health and population risk mapping. Identifying high-risk demographic groups allows authorities to allocate resources more effectively and design targeted awareness campaigns.

Despite the abundance of literature on Machine Learning-Based Heart Disease Diagnosis (MLBHDD), many existing studies rely on small, older datasets (such as the Cleveland dataset with only ~ 300 samples), which limits their generalization to broader populations. Furthermore, while diagnostic accuracy is critical, clinical applicability also demands model robustness—the ability to perform consistently on unseen data—and interpretability. Our project addresses these gaps by utilizing a large-scale dataset of 70,000 patients and implementing a rigorous pipeline that combines advanced feature engineering with state-of-the-art Gradient Boosting techniques.

2.2 Problem Definition

We formulate the cardiovascular disease prediction task as a supervised binary classification problem. Let D be a dataset of N patients, where x represents a vector of features (demographic and clinical variables for a patient), and y is the binary target variable.

- $y = 1$ indicates the presence of cardiovascular disease.
- $y = 0$ indicates the absence of disease.

Our goal is to learn a mapping function $f : X \rightarrow Y$ that predicts the probability of a patient having heart disease given their features $x \in X$, such that we maximize the classification performance on unseen data. Given the medical context, we aim to optimize not only Accuracy but also Recall (Sensitivity), as minimizing False Negatives (missing a sick patient) is crucial for patient safety.

3 Related Work

3.1 Traditional Cardiovascular Risk Prediction Models

Heart disease risk prediction systems have been developed and deployed for several decades, primarily in the form of rule-based or statistical risk scores tailored to specific populations or healthcare systems. Widely adopted models include SCORE in Europe, QRISK3 in the United Kingdom, the ASCVD Pooled Cohort Equations (PCE) in the United States, and the Framingham Risk Score (FRS). These models typically rely on a limited set of demographic and clinical variables such as age, sex, blood pressure, cholesterol levels, smoking status, and diabetes.

Despite their clinical utility, traditional risk scores suffer from several well-documented limitations. Most assume linear relationships between predictors and outcomes, restrict feature

space to manually selected variables, and are often optimized for specific endpoints (e.g., fatal events only) or populations, limiting generalizability. Table 1 summarizes the characteristics and major limitations of commonly used cardiovascular risk scores.

These constraints have motivated the exploration of data-driven approaches capable of modeling complex, non-linear interactions among heterogeneous risk factors.

Table 1: Characteristics and Limitations of Traditional Risk Scores

Risk Score	Geographic/Clinical Context	Primary Variables	Major Limitations
SCORE	Europe / Primary Care	Age, Sex, SBP, Total Cholesterol	Focuses only on fatal events[4]
QRISK3	UK / Primary Prevention	BMI, Ethnicity, Smoking, Deprivation	Overlooks heart failure[5].
ASCVD / PCE	USA / Statin Prescription	Age, BP, Diabetes, Smoking	Linear assumptions; limited features[4].
FRS	Global / Epidemiological	Lifestyle, Clinical markers, Genetics	Community-specific; lacks diversity [6].

3.2 Machine Learning for Heart Disease Diagnosis

Machine learning-based heart disease diagnosis (MLBHDD) has emerged as a promising alternative to conventional risk models due to its ability to capture non-linear relationships in high-dimensional data. In contrast to many engineering applications, research in medical ML prioritizes diagnostic accuracy and reliability over computational efficiency, as misclassification can have severe clinical consequences. Consequently, many studies emphasize minimizing classification error while preserving model interpretability [2, 3].

A large-scale systematic analysis by Liu et al. (2023), aggregating evidence from 16 studies comprising approximately 3.3 million patients, demonstrated that machine learning models consistently outperform traditional cardiovascular risk scores in terms of discrimination performance. Specifically, ML models achieved significantly higher C-statistics (the area under the curve for classification models), the standard, threshold-independent metric for cardiovascular risk prediction, indicating superior ability to distinguish high-risk from low-risk individuals across diverse populations [7]. Table 2 summarizes the comparative performance of major ML approaches relative to conventional risk scores.

While ensemble methods and neural networks generally achieve the highest discrimination, even classical ML models such as logistic regression and support vector machines demonstrate modest but statistically significant improvements over traditional scoring systems.

3.3 Deep Learning and Unstructured Clinical Data

The increasing availability of semi-structured and unstructured medical data—particularly medical images and physiological signals—has accelerated the adoption of deep learning methods in cardiovascular disease prediction. Convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid architectures are especially well-suited for learning representations directly from raw data such as electrocardiograms (ECGs).

Table 2: Comparative Performance of ML Approaches vs Traditional Scores

Method	ML C-statistic (95% CI)	Trad. Score C-statistic	P-value
Random forest	0.747 (0.684-0.809)	0.746	< 0.0001
Neural network	0.777 (0.722-0.832)	0.759	< 0.0001
Logistic regression	0.719 (0.633-0.805)	0.715	< 0.0001
SVM	0.709 (0.620-0.797)	0.711	0.0001
Other ML methods	0.778 (0.739-0.818)	0.772	< 0.0001

Ahmed and R (2025) compared artificial neural networks with several traditional machine learning models using the UCI Heart Disease dataset and reported that neural networks achieved the highest performance, with an accuracy of 97.67% and a precision of 100%, substantially outperforming Naïve Bayes and other baseline classifiers[8]. Similarly, Sun et al. (2024) proposed a CNN-LSTM-SE architecture in which CNN layers extract local spatial features from ECG signals, LSTM layers model temporal dependencies, and squeeze-and-excitation (SE) modules adaptively reweight informative channels. This model significantly outperformed classical approaches in arrhythmia classification, highlighting the advantage of deep learning in automatically learning spatial-temporal features from raw ECG data [9]. Comparable findings were reported by Obeidat and Alqudah (2021) using CNN-LSTM architectures for beat-wise ECG classification[10].

3.4 Benchmark Datasets

A substantial portion of the literature relies on publicly available benchmark datasets, such as the UCI Cleveland Heart Disease dataset and the MIT-BIH Arrhythmia dataset, due to their accessibility, standardized structure, and the ethical and regulatory constraints associated with collecting new medical data [3]. Additional datasets, including Framingham, Statlog, SPECTF, and Z-Alizadeh Sani, are frequently used to evaluate feature-based and population-level prediction models.

Table 3 summarizes the most commonly used datasets for cardiovascular disease detection, along with their dimensionality and typical research applications [11]. While these datasets facilitate reproducibility and benchmarking, many are limited in size, demographic diversity, or clinical scope, which may restrict the external validity of trained models.

Table 3: Benchmark Datasets for CVD Detection

Dataset	Features	Instances	Justification
Cleveland	76	303	Most cited; balanced benchmark.
Statlog	14	270	Compact; reduced dimensionality.
Framingham	15	4133	Large-scale; population-based.
SPECTF	44	267	Feature-based analysis.
Z-Alizadeh Sani	54	303	Clinical/echocardiographic data.

3.5 Model Interpretability and Explainable AI

Despite their predictive strength, deep learning and ensemble models are often criticized for their lack of transparency. In clinical settings, interpretability is essential, as clinicians must

understand the rationale behind a model’s prediction before integrating it into diagnostic or treatment decisions. This has led to growing interest in explainable artificial intelligence (XAI) techniques for cardiovascular risk prediction [12].

SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) are among the most widely adopted XAI methods. SHAP, grounded in cooperative game theory, quantifies the contribution of each feature to a model’s prediction relative to a baseline expectation. In heart disease prediction tasks, SHAP analyses consistently identify age, cholesterol levels, blood pressure, and family history as dominant contributors to predicted risk [13, 14]. LIME, in contrast, focuses on local interpretability by approximating the model’s behavior in the neighborhood of an individual instance, enabling clinicians to inspect patient-specific drivers of risk, such as ECG abnormalities or elevated biomarkers. [12, 15]

Together, these methods play a critical role in increasing trust, transparency, and clinical acceptance of machine learning-based cardiovascular decision support systems.

4 Methodology

This study frames the prediction of cardiovascular disease as a binary classification problem. The target variable, *cardio*, indicates the presence (1) or absence (0) of cardiovascular disease. Formally, the prediction task is defined as learning a function:

$$f : X \rightarrow Y \quad (1)$$

where $X = \{x_1, x_2, \dots, x_n\}$ represents the feature space of patient attributes, and $Y \in \{0, 1\}$ denotes the binary outcome.

4.1 Dataset Description

We employ the Cardiovascular Disease Dataset (2019), containing medical records for 70,000 patients, publicly available on Kaggle. The features comprise continuous measurements, ordinal indicators, and binary lifestyle factors.

Table 4: Dataset Variables

Variable	Description	Type
age	Age (days)	Numeric
height	Height (cm)	Numeric
weight	Weight (kg)	Numeric
gender	1 = Female, 2 = Male	Categorical
ap_hi	Systolic Blood Pressure	Numeric
ap_lo	Diastolic Blood Pressure	Numeric
cholesterol	1: Normal, 2: Above Normal, 3: High	Ordinal
gluc	1: Normal, 2: Above Normal, 3: High	Ordinal
smoke	Smoking (0/1)	Binary
alco	Alcohol Intake (0/1)	Binary
active	Physical Activity (0/1)	Binary
cardio	Target: Disease Presence (0/1)	Binary

4.2 Exploratory Data Analysis

Initial inspection confirmed no missing values. Descriptive statistics revealed extreme outliers in blood pressure measurements (e.g., negative values or values > 10000), likely due to data entry errors. The target variable (*cardio*) is near-balanced, reducing concerns about class imbalance.

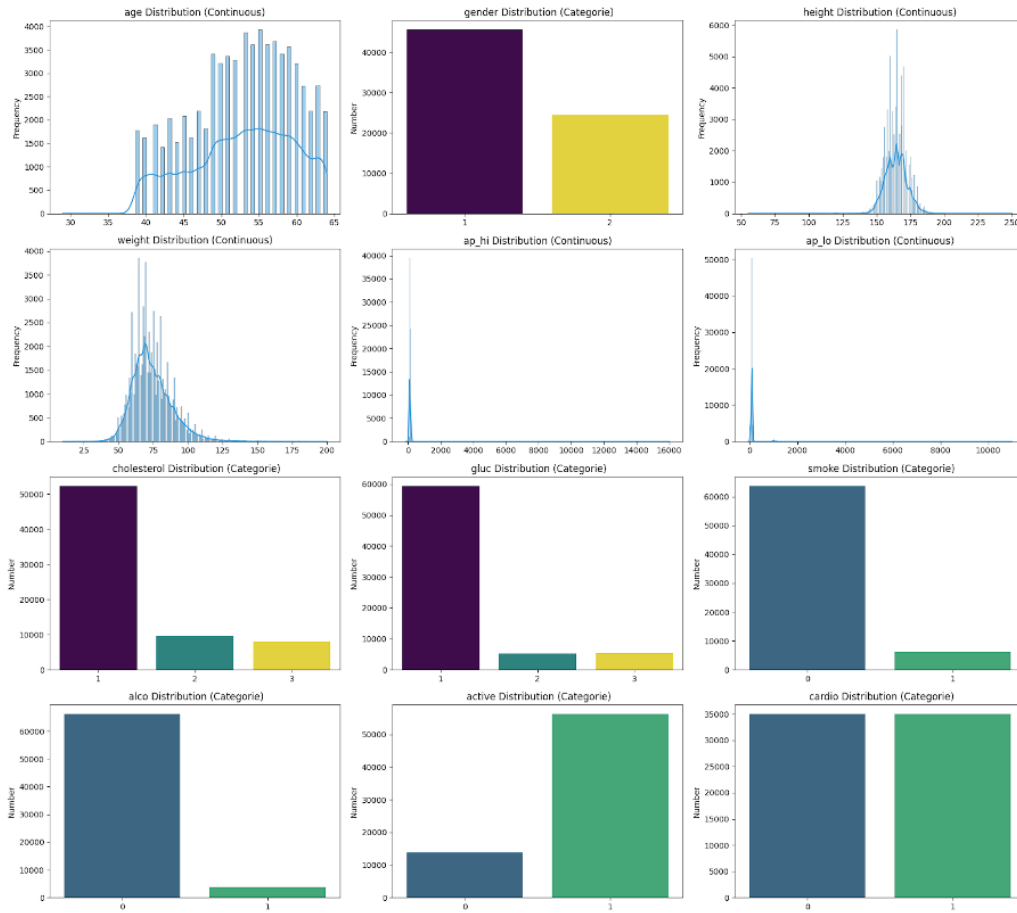


Figure 1: Distributions of continuous and categorical variables (Age, Weight, Cholesterol, etc.).

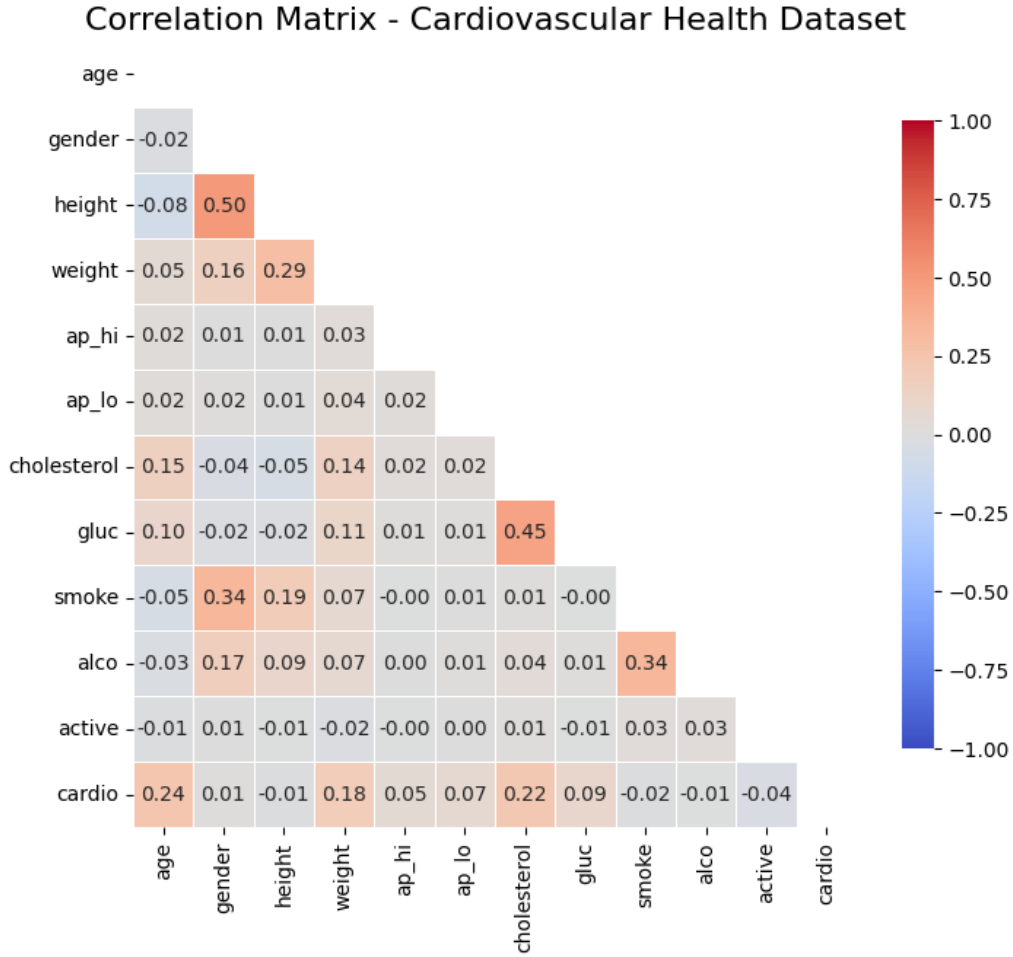


Figure 2: Pearson correlation matrix of the unprocessed dataset.

4.3 Data Cleaning and Preprocessing

Age values were converted from days to years. Outlier correction focused on systolic (*ap_hi*) and diastolic (*ap_lo*) blood pressure[16, 17].

- Systolic BP constrained to [50, 250] mmHg. [18, 19]
- Diastolic BP constrained to [30, 160] mmHg [18, 19].
- Records where diastolic pressure exceeded systolic were removed [20, 21].
- 163 anomalous observations were excluded.

Continuous variables were standardized using Z-score normalization.

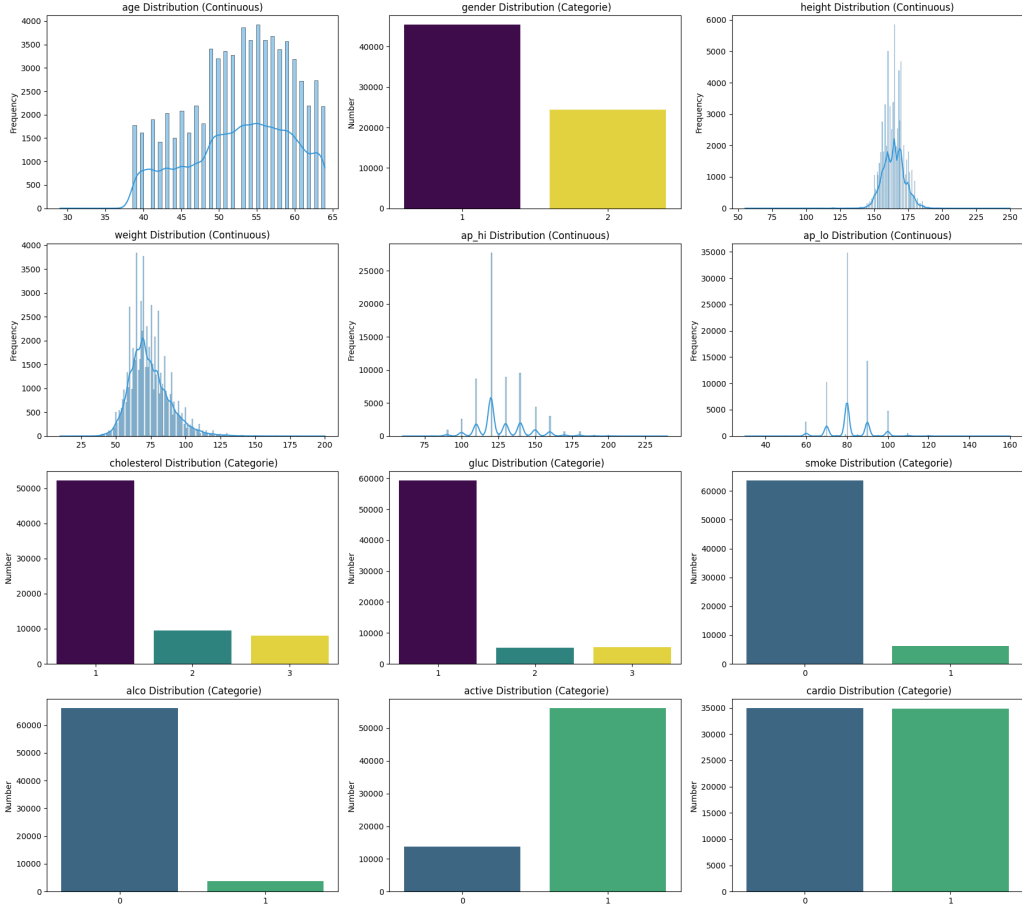


Figure 3: Distributions of processed continuous and categorical variables (Age, Weight, Cholesterol, etc.).

4.4 Feature Engineering

To capture clinically meaningful patterns, we derived the following features:

1. **Body Mass Index (BMI)** [22]:

$$BMI = \frac{\text{weight (kg)}}{(\text{height (m)})^2} \quad (2)$$

2. **Mean Arterial Pressure (MAP)**[23] averages pressure through a cardiac cycle.

$$MAP = \frac{ap_hi + 2 \times ap_lo}{3} \quad (3)$$

3. **Clinical Categorization:** BMI was binned into 6 levels (Underweight to Morbid Obesity [24]), and Blood Pressure into 5 stages (Normal to Hypertensive Crisis [25]).

4. **Aggregate Risk Scoring:** Composite features were created to summarize cumulative patient risk:

- **Lifestyle Risk Score:** A binary additive indicator combining smoking and alcohol consumption [26].
- **Metabolic Risk Score:** A count-based feature (0–4) representing the presence of obesity, hypertension, hyperglycemia, and hypercholesterolemia [27, 28].

5 Evaluation

5.1 Experimental Setup

To ensure a rigorous evaluation and avoid data leakage, we applied the following protocol:

- **Data splitting:** After cleaning, the dataset contains $N = 69,837$ observations and was split into a training set (80%) and a testing set (20%) using stratified sampling to preserve the class proportions.
- **Cross-validation:** During model selection and hyperparameter tuning, we used a 10-fold cross-validation strategy on the training set.
- **Metrics:** In this medical screening context, we report Accuracy for overall performance and prioritize Recall (Sensitivity) and F1-score to limit False Negatives.

5.2 Evaluation Metrics

We evaluate models using:

- **Accuracy:** $\frac{TP+TN}{TP+FP+TN+FN}$
- **Precision:** $\frac{TP}{TP+FP}$
- **Recall:** $\frac{TP}{TP+FN}$
- **F1-Score:** $\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

6 Results and Discussion

6.1 Model Comparison and Selection

We experimented with three distinct families of algorithms: Logistic Regression (linear), K-Nearest Neighbors (distance-based), and XGBoost (ensemble tree-based).

- **Logistic Regression:** Accuracy $\approx 71.5\%$. While interpretable, it struggles with non-linear relationships.
- **KNN:** Performance plateaued around $\approx 69\%$, impacted by the curse of dimensionality and high inference cost.
- **SVM:** Performed extremely and was the closest to XGBoost with an Accuracy score $\approx 73\%$.
- **XGBoost:** Outperformed baselines due to its ability to model non-linear interactions and its robustness.

We compared ROC curves across models to evaluate discrimination across all thresholds.

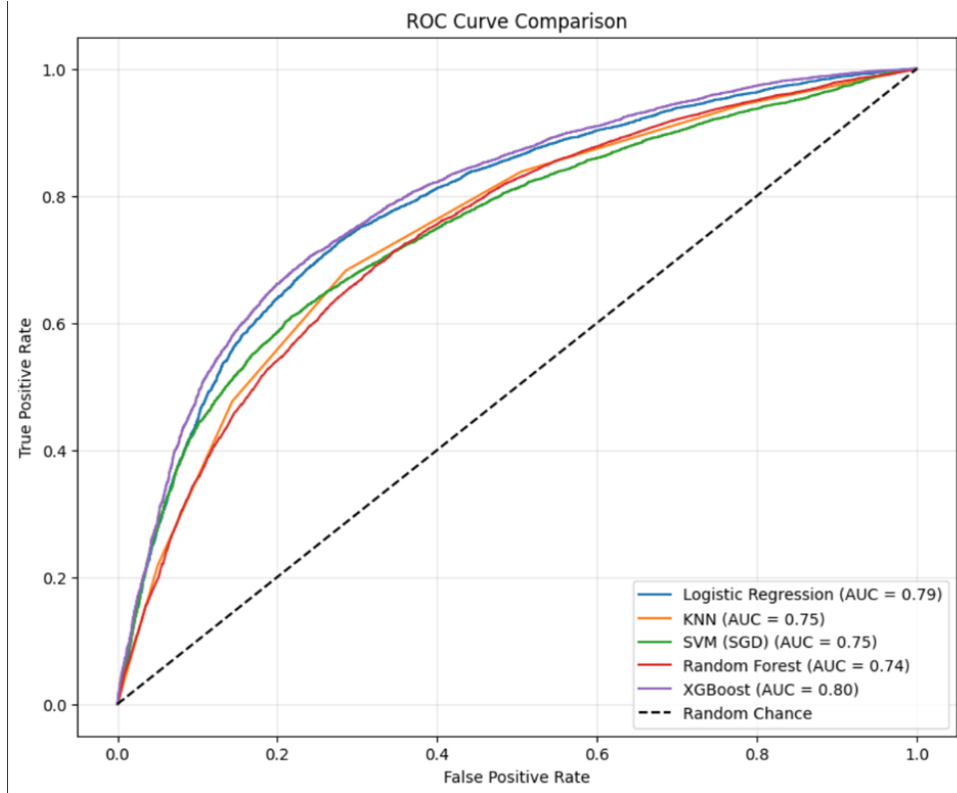


Figure 4: ROC curves for multiple algorithms (XGBoost achieves the highest AUC).

XGBoost achieved the best AUC (**0.80**), and stays consistently above the others especially in the high sensitivity part (left side of the graph) which indicates a superior True Positive Rate for a given False Positive Rate. Logistic Regression result (**0.79**) suggests that linear relationships have an important impact for the prediction of this model. However, it fails to capture complex non-linear relationships that allowed XGBoost to reach optimal performance, while KNN and Random Forest were lower (around **0.74–0.75**).

After Grid Search optimization, the best XGBoost model was configured with:

$$n_estimators = 300, \quad learning_rate = 0.05, \quad max_depth = 5, \quad subsample = 0.8$$

6.2 Quantitative Results

Table 5 reports the performance of the best model on the unseen test set.

Table 5: Performance metrics of the best XGBoost model.

Metric	Class 0 (Healthy)	Class 1 (Disease)	Weighted Avg.
Precision	0.72	0.74	0.73
Recall	0.76	0.71	0.73
F1-score	0.74	0.72	0.73
Accuracy	–	–	73.15%

To verify robustness, we compared training vs testing accuracy. The training accuracy was 75.31% and the test accuracy 73.15%, yielding a small generalization gap of 2.16%.

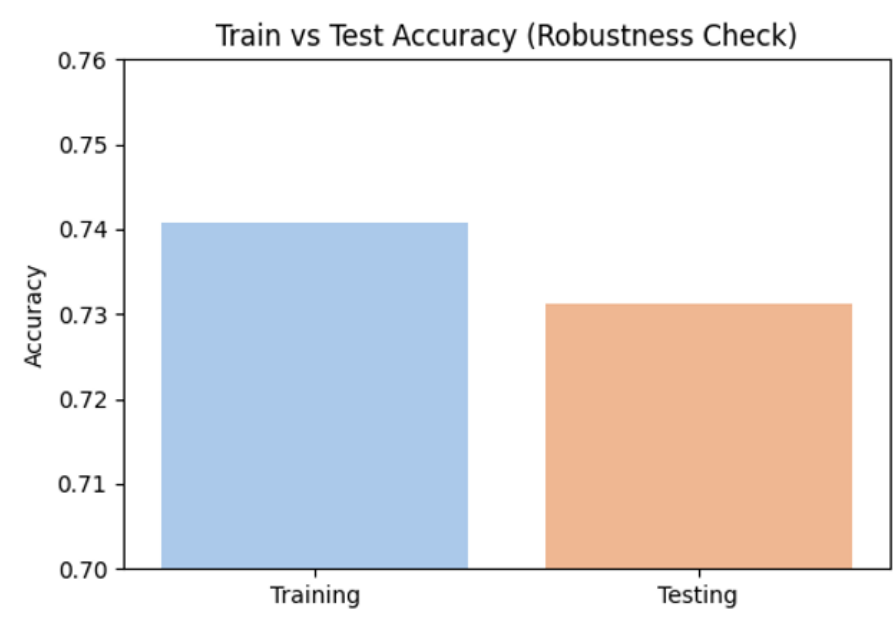


Figure 5: Training vs testing accuracy (generalization gap).

6.3 Error Analysis (Confusion Matrix)

To better understand limitations, we analyzed classification errors using the confusion matrix.

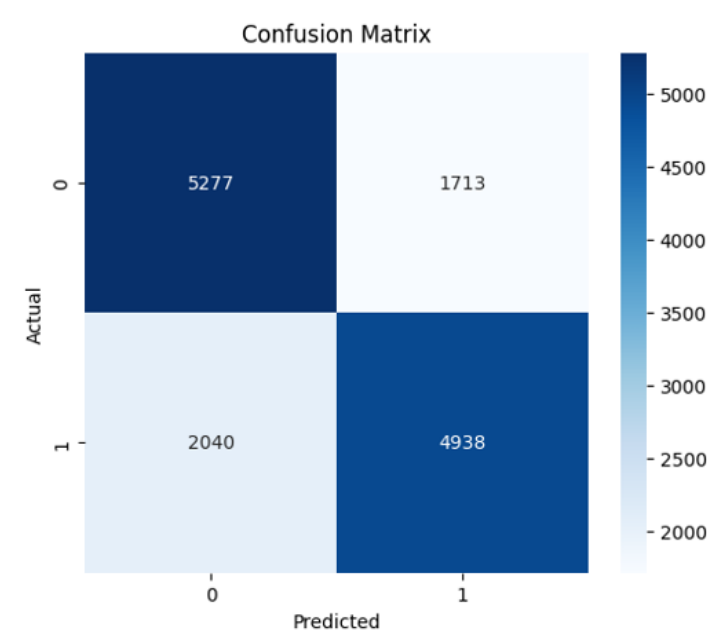


Figure 6: Confusion matrix of the best XGBoost model (threshold=0.5).

- **True Positives:** The model correctly identifies about 71% of diseased patients.
- **False Negatives (Type II error):** About 29% of positive cases are missed, which is critical in screening settings.
- **False Positives (Type I error):** About 24% of healthy patients are flagged as at-risk; in screening, this is often acceptable (additional testing rather than missed diagnosis).

6.4 Fairness Analysis

We briefly evaluated performance across gender subgroups:

- **Male patients:** Accuracy $\approx 72\%$
- **Female patients:** Accuracy $\approx 74\%$

The model performs slightly better on female patients, yet the gap is small, suggesting comparable performance across genders on this dataset.

6.5 Advanced Interpretability (SHAP Analysis)

XGBoost feature importance indicates which variables matter but not the direction of their effect, so we complemented it with SHAP.

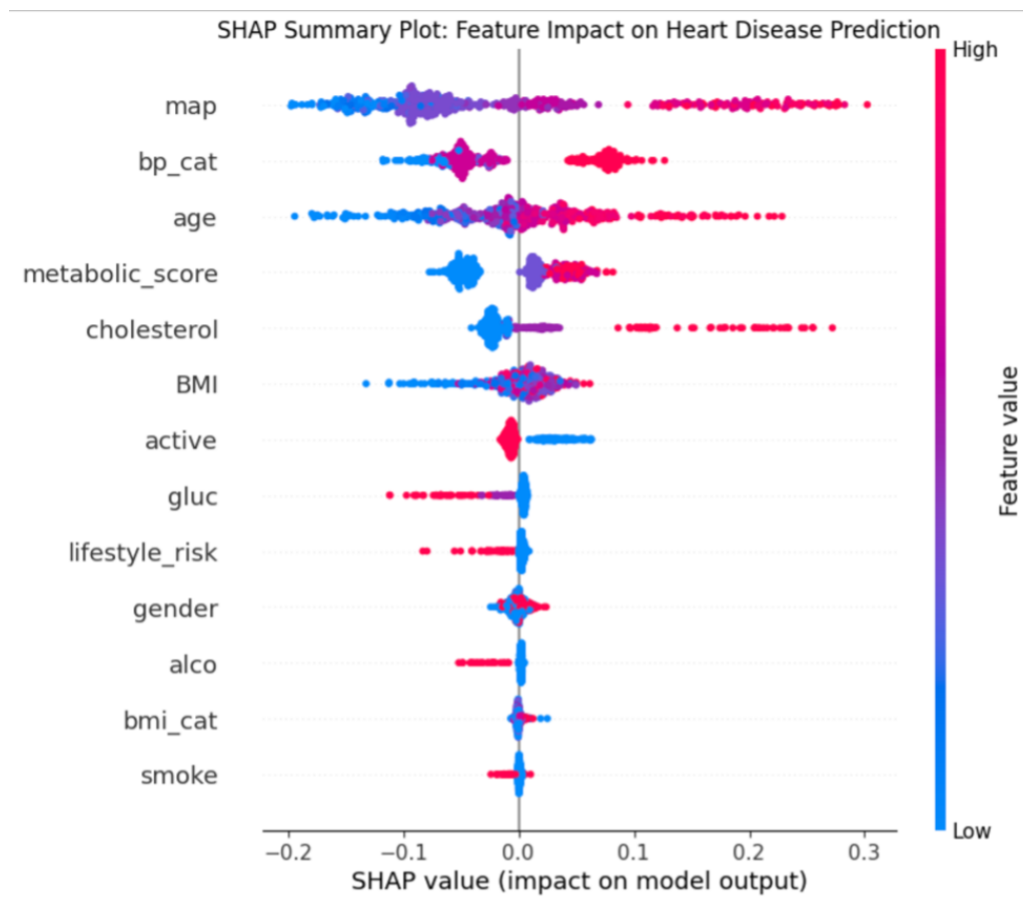


Figure 7: SHAP summary plot for the best XGBoost model.

Interpretation highlights:

- Blood pressure features (e.g., *map*, *bp_cat*) are dominant contributors. The Mean Arterial Pressure (*map*) and Blood Pressure Category (*bp_cat*) are the two top contributors to the model's decisions. Mean Arterial Pressure is wide spread along the X-axis and shows a high discriminatory power: high values represented in red dots are strongly condensed on the right positive side, while the low values are mostly displayed on the other side (left negative). This means that high values increase the probability of the prediction being positive, whereas low values increase the probability of the prediction being negative.

- Engineered variables (e.g., *map*, *bp_cat*, *metabolic_score*) appear among the top features, supporting the relevance of feature engineering.
- This analysis helps to underlie the protective factors by capturing the inverse relationship. For instance, the ‘active’ variable’s high values (active people) are clustered on the left side while the low values (non-active people) are concentrated on the right side. This shows that physical activity is interpreted as a protective factor that reduces the probability of disease, whereas inactivity shifts the prediction towards higher risk..

6.6 Feature Correlation Analysis

To validate feature relevance, we computed a Pearson correlation matrix on the processed dataset.

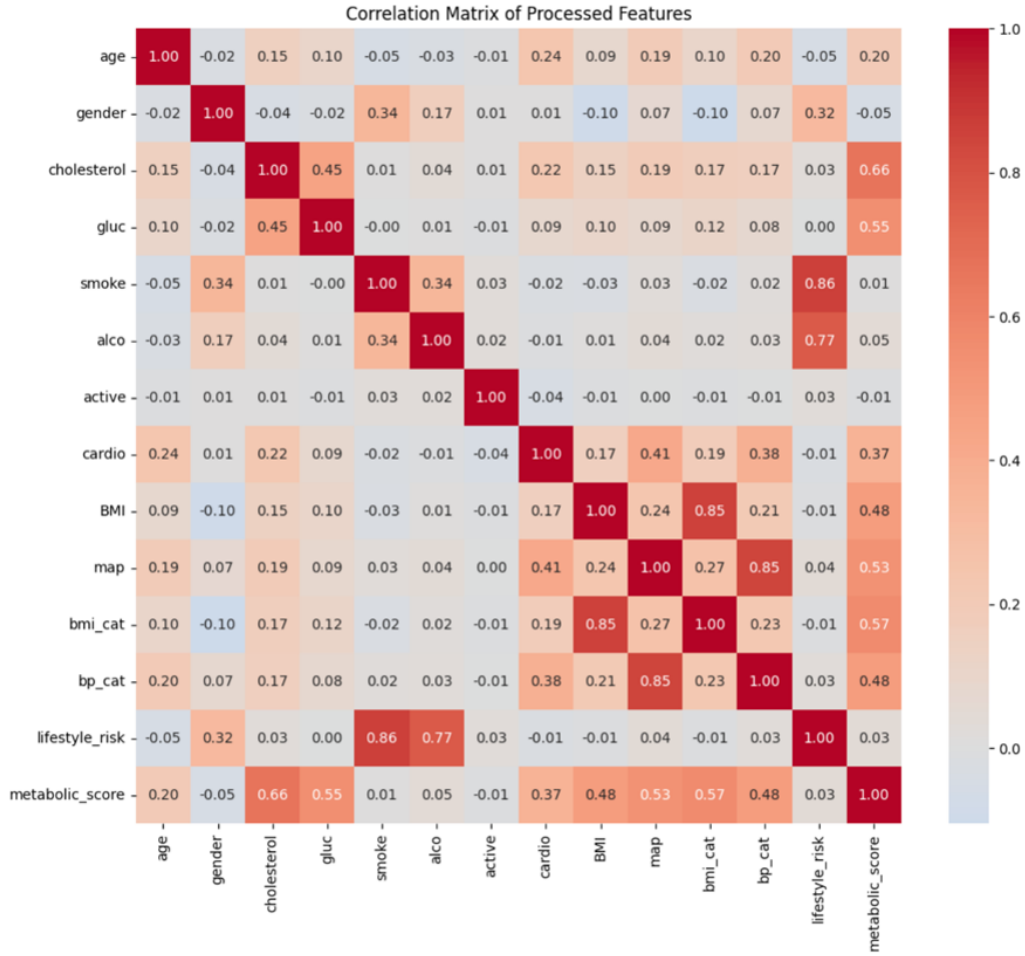


Figure 8: Pearson correlation matrix of the processed dataset.

Notable correlations with the target (*cardio*) include *map*: $r = 0.41$, *bp_cat*: $r = 0.38$, *metabolic_score*: $r = 0.37$ in line with the SHAP analysis confirming that blood pressure metrics are the most relevant features to predict heart disease.

The analysis also confirms the choice of the model (XGBoost). We can observe that there are strong correlations between continuous variables and their categorical counterparts (*BMI* vs *bmi_cat*, $r = 0.85$). This situation can deteriorate a linear model, our choice of XGBoost (tree-based model) allows us to retain both representations as it handles redundant features by selecting the optimal split at each node.

Additionally, it confirms the consistency of Composite Scores. The engineered feature *lifestyle_risk* shows a strong correlation with *smoke* ($r = 0.86$) and *alco* ($r = 0.77$), which validates that behavioral risk factors were successfully aggregated into a single dimension.

6.7 Clinical Safety and Threshold Calibration

With the default decision threshold (0.5), recall is about 71%, leaving many positives undetected. We therefore searched for an operational threshold that reduces false negatives while keeping acceptable accuracy.

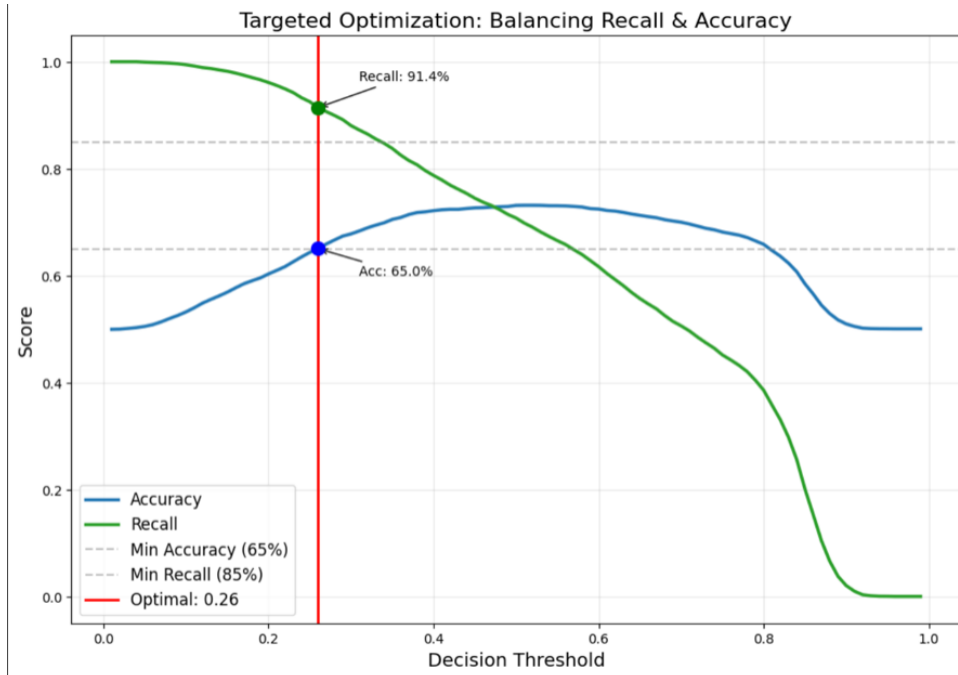


Figure 9: Threshold calibration: accuracy/recall trade-off vs decision threshold.

We selected an optimal threshold of **0.26**. This increased Recall to **91.4%** and reduced False Negatives from 2,040 to 598, while keeping Accuracy at **65.0%**. It ensures that our tool remains credible for medical usage and avoids the “false alarm fatigue” by lowering the threshold even more.

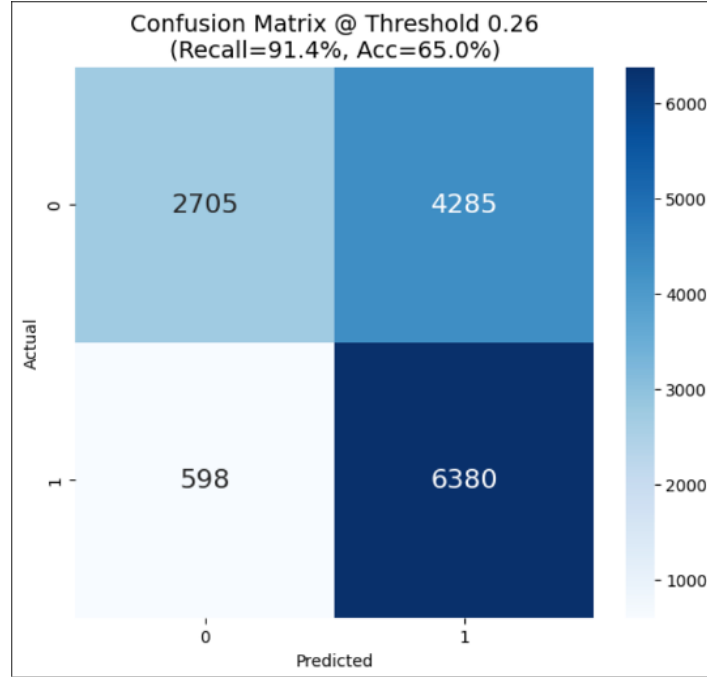


Figure 10: Confusion matrix after threshold calibration (threshold=0.26).

7 Conclusion

Our study demonstrates that tree-based ensemble methods like XGBoost, combined with domain-specific feature engineering, can effectively predict cardiovascular risk. Systolic blood pressure remains the dominant predictor. Model performance aligns with results reported on widely used benchmark datasets, though clinical usability remains challenged by the accuracy loss required to prioritize recall. Future efforts will aim to reduce false negatives and identify a minimal set of easily measurable features to strengthen clinical safety and support operational deployment.

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